MILITARY SPECIFICATION

CATHETER, UMBILICAL, POLYVINYL CHLORIDE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes of sterile, disposable, umbilical artery or vein catheters suitable for measuring blood pressure, doing blood chemistry and blood volume studies, intravenous infusions and exchange transfusions of newborn infants.

1.2 Classification. Catheters shall be furnished in the following sizes, as indicated below (see 6.1):

Sizes French (Fr)

A - 3-1/2
B - 5
C - 8

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water-Resistant, (For Carton Sealing).

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Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet, overwrap; Water-vaporproof or Waterproof, Flexible.
MIL-S-36586	Sterilization Strip Test, Bacterial Spore.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pa. 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-412	Method of Tension Testing of Vulcanized Rubber.
ASTM D-2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, Pa. 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1 (UDC 615.47)	American Standard Dimensions for Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Material. Catheters shall be fabricated from transparent, clear, pliable, medical grade plastic, suitable in all respects for the purpose intended. Shall be thermosensitive polyvinyl chloride plastic which yields easily to tissue contours at body temperature. Plastic shall meet the requirements of The Pharmocopeia of the United States of America (USP) for Class VI plastic.

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3.2 Style, design and dimensions. Shall be a sterile, nonpyrogenic, nontoxic, disposable, flexible tube with full length radiopaque stripe. The proximal end shall be funnel-shaped and form a tight, leakproof fit with a male Luer connector, meeting ANSI Z70.1, on a 3-way stop cock. The distal end shall have an open tip with no lateral eyes. Catheter shall have a transverse permanent mark at distances of 5, 10 and 15 centimeters from the distal end. Shall be smooth inside and out. Shall be in accordance with Figure 1 for Size A, Size B or Size C, as appropriate.

3.3 Sterility. The umbilical catheters shall be sterile when tested with a biological indicator as specified in 4.4.4.

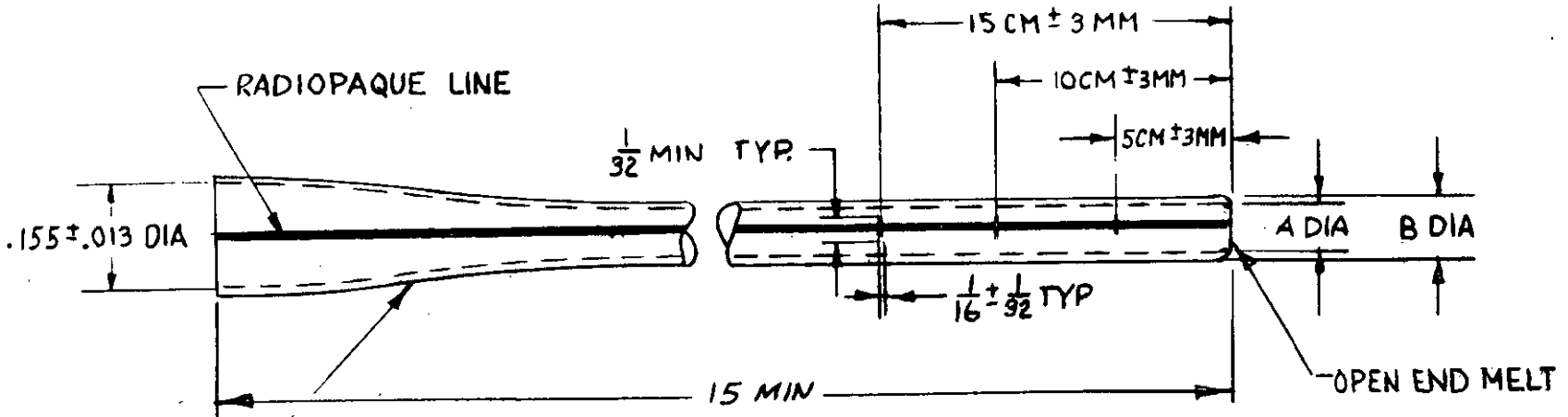
3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. Federal stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

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DIMENSIONS ARE IN INCHES, UNLESS OTHERWISE STATED.

NSN	SIZE	DIMENSIONS	A ±.013	B ±.013
6515-00-360-5162	A	3½ FR.	.025	.043
6515-00-133-5877	B	5 FR.	.037	.063
6515-00-360-5163	C	8 FR.	068	.102

CATHETER, UMBILICAL, POLYVINYL CHLORIDE, DISPOSABLE

FIGURE 1

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3.4 Pyrogens. The umbilical catheters shall be non-pyrogenic when tested as specified in 4.4.5.

3.5 Safety (non-toxic). The umbilical catheters shall be safe (non-toxic) when tested as specified in 4.4.6.

3.6 Animal implantation test. The umbilical catheters shall produce no toxicity or adverse tissue reaction evident upon gross and microscopic examination, when tested as specified in 4.4.7.

3.7 Certificate of freedom from pyrogens, safety and animal implantation tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be free from pyrogens, safe (non toxic) and have passed the animal implantation test. This certificate shall include the specific tests performed and the test results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical, Materiel, 2800 South 20th Street, Philadelphia, Pa., 19101, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.8 Compatibility. The materials of which the catheters are manufactured shall not react with, or be acted upon, by the liquids with which they will be in contact during normal use, nor shall there be any materials incorporated in the catheter which will cause any harmful or deleterious effect when the tube is used for the purpose intended.

3.9 Radiopacity. When tested as specified in 4.4.2, catheter shall throw a definite uniform shadow. The degree of opaqueness shall be such that the shadow of a small foreign body in the front or rear of the catheter shall be visible for diagnosis.

3.10 Breaking strength. The breaking strength of the catheters shall be 2.5 lbs, 6.0 lbs. and 8.0 lbs., minimum, for sizes A, B and C, respectively, when tested as specified in 4.4.1.

3.11 Hardness. Catheters shall have a Shore Durometer Hardness of 85 $\pm$ 3 (Shore A, 10 seconds) when tested as specified in 4.4.3.

3.12 Leakage. Shall not leak when tested for leakage in accordance with 4.4.8.

3.13 Workmanship. Workmanship shall be first class throughout. Umbilical catheters shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one catheter.

Table I. Sampling for Examiantion

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for test purposes shall be one catheter.

Table II. Sampling for Tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Breaking strength	3.9	4.4.1	S-1	*
Radiopacity	3.8	4.4.2	S-1	*
Hardness	3.10	4.4.3	S-1	*
Sterility	3.3	4.4.4	4.4.4	*
Leakage	3.12	4.4.8	S-1	*

* Acceptance Number shall be zero for all sample sizes.

4.2.3 Test samples. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

4.3 Examination. The umbilical catheters shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any umbilical catheter in the sample, containing one or more defects shall be rejected, and if the number of defective umbilical catheters in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Not free of roughness at distal tip (not melt smooth).
102	Style or design not as indicated in 3.2 (see Figure 1).
103	Catheter kinked or otherwise obstructed.
104	Presence of foreign matter on exterior or interior of catheter tube.
105	Distance markings incomplete, missing or not distinct.
Minor	
201	Distance markings not square or rectangular in shape.
202	Exterior surface not smooth.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Breaking strength. Breaking strength shall be determined in accordance with ASTM D 412, Standard Method of Tension Testing of Vulcanized Rubber.

4.4.2 Radiopacity. Each sample of an umbilical catheter shall be radiographed as follows:

85KVP, 10 milliampere-seconds, 25 inch distance, 24 centimeters of paraffin USP superimposed over the tube and using a standard make of cassette with screens, films and processing solutions. The foreign body used for test purposes shall be a small fragment of sheet lead 1/2 millimeters in thickness.

4.4.3 Hardness. Shore "A" Durometer Hardness shall be determined as specified in ASTM D-2240, Standard Method for Test of Indentation Hardness of Rubber and Plastics by Means of a Durometer.

4.4.4 Sterility test. The umbilical catheters shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.4.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units.

4.4.4.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.4 Sterility test and 4.4.4.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.4.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.4.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.5 Pyrogens. Umbilical catheters shall be tested for freedom from pyrogens in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.6 Safety (non-toxicity). Umbilical catheters shall be tested for safety (non-toxicity) in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.7 Animal Implantation test. Samples of the umbilical catheters shall be tested in accordance with the method given in the USP for Implantation Test, except for the following changes: Last paragraph, second line, after "72 hours" add "not to exceed 144 hours." Last paragraph, delete third and fourth sentences and substitute: "The tissues surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or The American College of Veterinary Pathologists."

4.4.8 Leakage test. Attach flared end of tube to male Luer fitting (in accordance with ANSI Z70.1). Knot distal portion of catheter and apply 45 (psig) hydraulic pressure for 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing 10 catheters, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. Each catheter shall be packaged in a sealed peel type package. The package shall be capable of maintaining sterility of contents unless damaged or opened.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistatn, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate container. Each immediate container shall be marked with the item name, lot (control) number, name and address of the manufacturer and the statement: "STERILITY VOID IF PACKAGE BROKEN" or similar.

5.3.2 Unit and intermediate package. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization (G-Gas, R-Radiation, S-Steam and D-Dry Heat) shall be shown following the level of packaging and date of packaging. Example: "A-4/74-G." Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Intended use. The umbilical catheters are intended for use in obtaining access to the umbilical artery or vein of infants in diagnosis and treatment involving blood and medications.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-133-5877	CATHETER, UMBILICAL, Polyvinyl Chloride, Disposable, 5 Fr, 10s
6515-00-360-5162	CATHETER, UMBILICAL, Polyvinyl Chloride, Disposable, 3-1/2 Fr, 10s
6515-00-360-5163	CATHETER, UMBILICAL, Polyvinyl Chloride, Disposable, 8 Fr, 10s

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1844

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at *reasonable cost and minimum delay*, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. *Comments submitted on this form do not* constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an *envelope addressed to preparing activity.*

DOCUMENT IDENTIFIER AND TITLE

MIL-C-37063

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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